

CSCI 5302

Advanced Robotics

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (50.8 / 100)	33.0 / 100

33	83	+69
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Exposes students to current research topics in the field of robotics and provides hands-on experience in solving a grand challenge program.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	50.8	72.8	+22.0	MEDIUM
Applied Scientist	35.1	99.2	+64.0	HIGH
Machine Learning Engineer	33.2	67.7	+34.5	HIGH
Data Scientist	31.0	100.0	+69.0	HIGH
ML Engineer	26.4	77.7	+51.2	HIGH
MLOps Engineer	21.3	82.0	+60.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (

example

) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

AI Engineer

50.8 → 72.8 (+22.0 pts)

perception data models learning software technical model systems

Applied Scientist

35.1 → 99.2 (+64.0 pts)

lab computational reasoning literature models technical data learning

Machine Learning Engineer

33.2 → 67.7 (+34.5 pts)

data learning tools exploratory design development science models

Data Scientist

31.0 → 100.0 (+69.0 pts)

data presentation different analytical people

ML Engineer

26.4 → 77.7 (+51.2 pts)

data learning systems models model world impact new

MLOps Engineer

21.3 → 82.0 (+60.7 pts)

detection data model scientists systems learning image training

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.